

Set Point Theory

What is Set Point?

Many parts of our physical and psychological makeup are determined by our genes. For example, height is mostly determined by genetic factors - some environmental factors may influence it a little, but for the most part, it is 'set'. Some people are shorter than average while others are taller than average. Most people understand and accept that our height is something we can't change or control; it's just the way we were born.

In the long term, the same principle applies to weight. Genetics play a large part in determining the weight that our bodies are most comfortable at, and this depends on our bone structure, metabolism, muscle composition, and other traits.



Research suggests that every human body has a natural weight range that it is in part determined by genetics. This weight range is where your body is in balance and is called your "set point". Set point will vary for every individual regardless of other factors such as their height and sex.

Set Point Theory and Body Weight

The human body has built in ways to regulate weight within this natural set point range. For example, if you eat a little more than you need to maintain your body weight, then typically your body temperature will rise and your metabolism speeds up to burn off the extra energy. If, on the other hand, you do not eat enough to maintain your weight, then your metabolism slows down to hold onto the available energy (see our handout on [Metabolism and Eating Disorders](#) for more info). Another way your body regulates is hunger - if your body is not getting enough energy, you may feel more hungry and/or be more preoccupied with food.

Consider someone who typically eats a balanced diet and exercises moderately. For much of their adulthood, their weight will likely fluctuate within their set point weight range. Their weight may go up temporarily if they go on a holiday and eat more energy dense foods. However, upon their return home, once they resume their normal pattern of eating and exercise, their weight will naturally go down again without effort. A person's weight may also temporarily go down if they become unwell. Once they recover, the body will then fight to regain the lost weight by increasing hunger levels and slowing their metabolism temporarily.

Set point is particularly important to understand if you have an eating disorder. You may be engaging in periods of restricted eating, and if this results in you falling below your body's natural set point weight range, your body may respond by increasing your hunger signals (at least initially), increasing your mental preoccupation with food, and decreasing your metabolism (see our handout on [Starvation Syndrome](#) for more information). These mechanisms are in place to try and get your body back to its natural weight range. They were developed through evolution to protect us from starvation.

Trying to keep your body below its natural set point weight range leads into an ever-increasing battle with these mechanisms. Preoccupation with food becomes so overwhelming that it can be hard to concentrate on anything else. Due to increased levels of hunger and increased thoughts about food, restricting your food intake makes you vulnerable to episodes of over eating or binge eating. This is why dieting is so ineffective (see our handout on [Why Diets Do Not Work](#) for more info).

Eating a balanced diet and staying active in ways that suit your lifestyle will support your overall health and wellbeing, and help your body maintain a weight range that is natural and sustainable for you.

If you are experiencing food preoccupation due to restricted food intake, the key lesson to learn from set-point theory is this: *To recover from your preoccupation with food, weight, and hunger, your body must be returned to a weight that is within its natural set point. No amount of psychological treatment will remove your food obsessions unless weight is regained to a healthy, normal range for your body.*

Accepting Your Set Point

Perhaps your set point weight range is higher than average, higher than you'd like it to be, or higher than your doctor, peers, or the media have suggested it 'should' be. In a world increasingly focussed on eating and size, this can feel very distressing. You can work with a psychologist to explore body acceptance and navigating the challenges of diet culture. Body acceptance *does not* mean loving your body at all times, at every angle, in every outfit. It is about helping you ensure that your body image concerns do not get in the way of living a life you value (see our modules [Caring Less About Your Looks](#) for more info).

